Fibonacci over powers of 2

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Let (a_n) be the Fibonacci sequence defined by $a_1 = a_2 = 1$ and $a_{n+1} = a_n + a_{n-1}$ for $n > 1$. Show that

$$\frac{a_1}{2} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \cdots + \frac{a_n}{2^n} < 2.$$